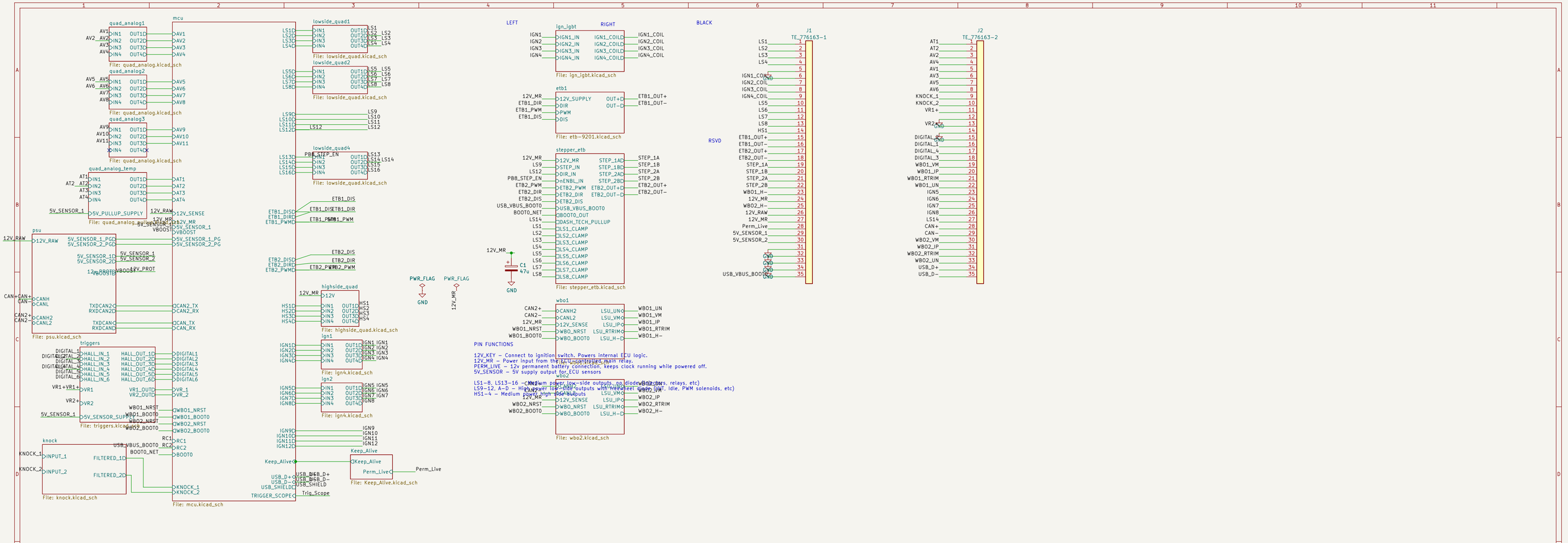
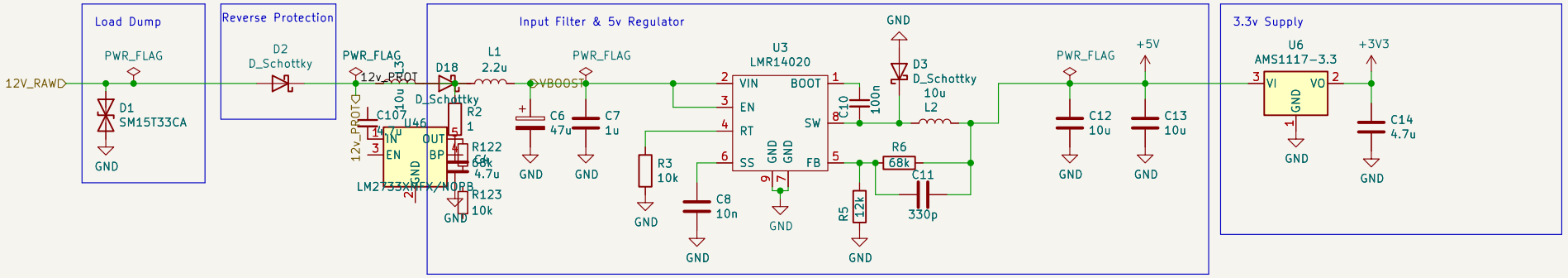
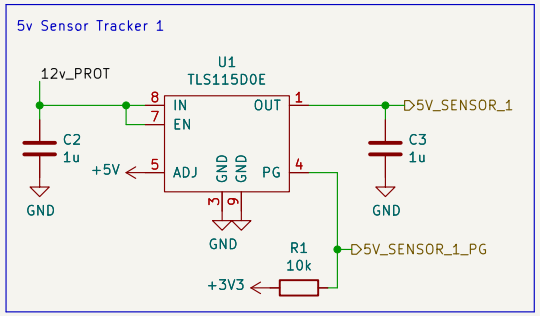
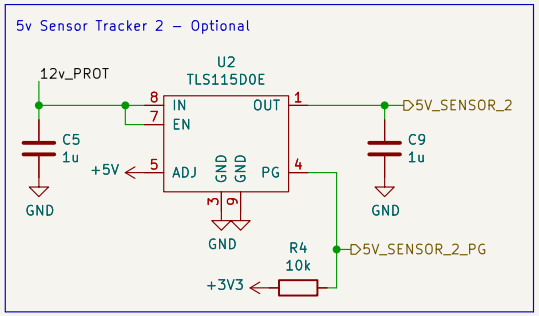
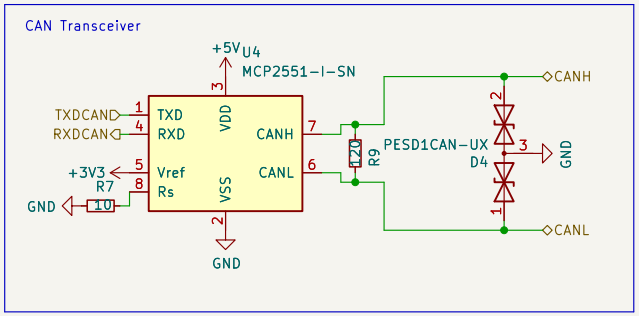


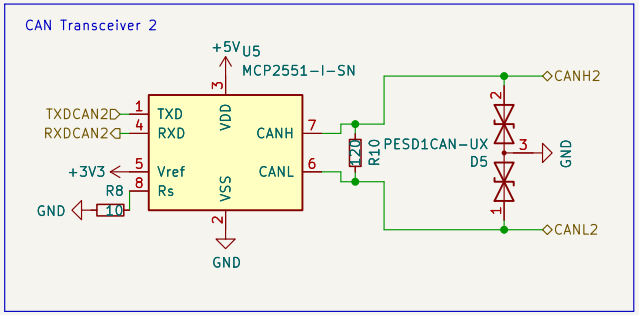
Sheet: /		
File: fome-mc.kicad_sch		
Title: fome-mc Integrated Motorcycle ECU		
Size: A2	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 1/21









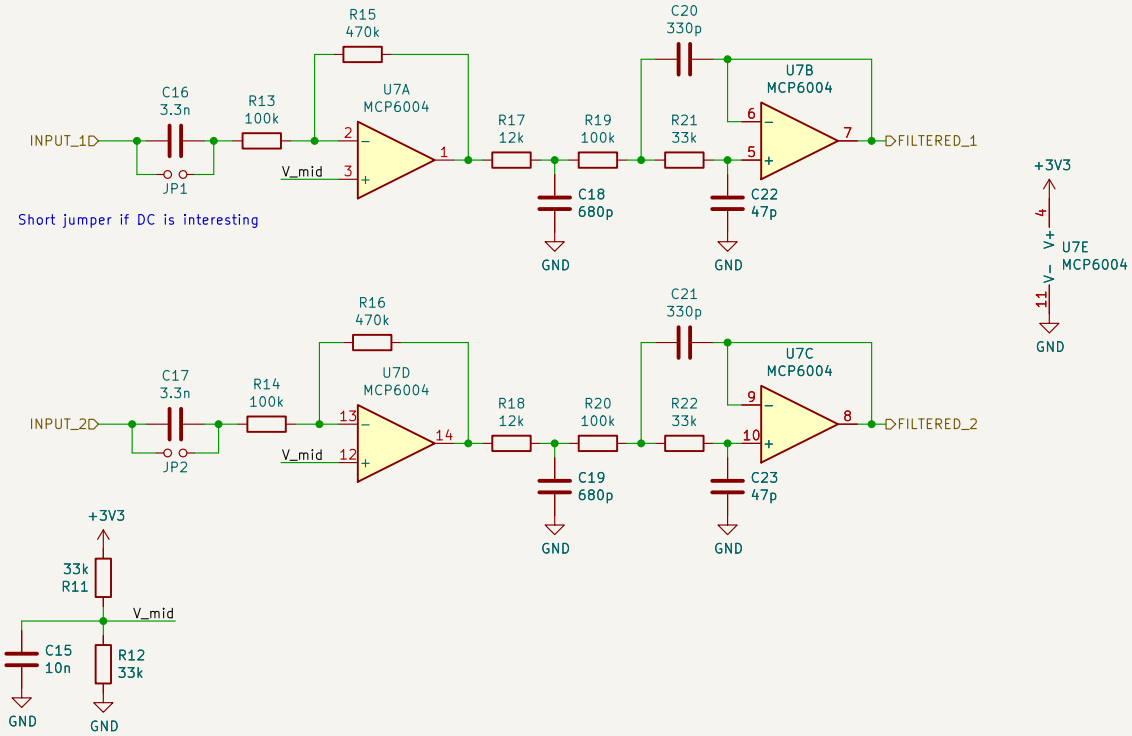


rusefi.com/s/proteus

Sheet: /psu/
File: psu.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 2/21

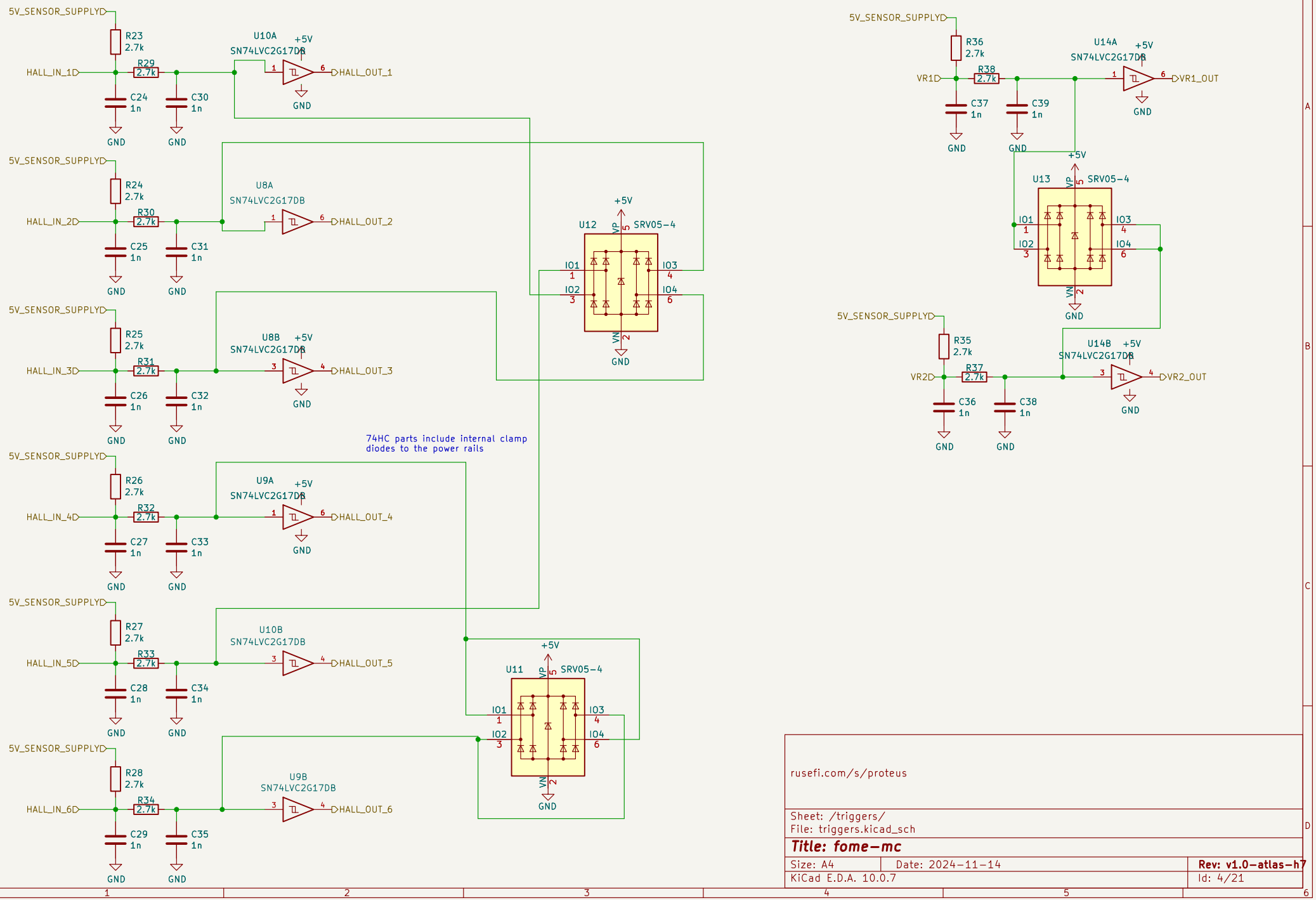


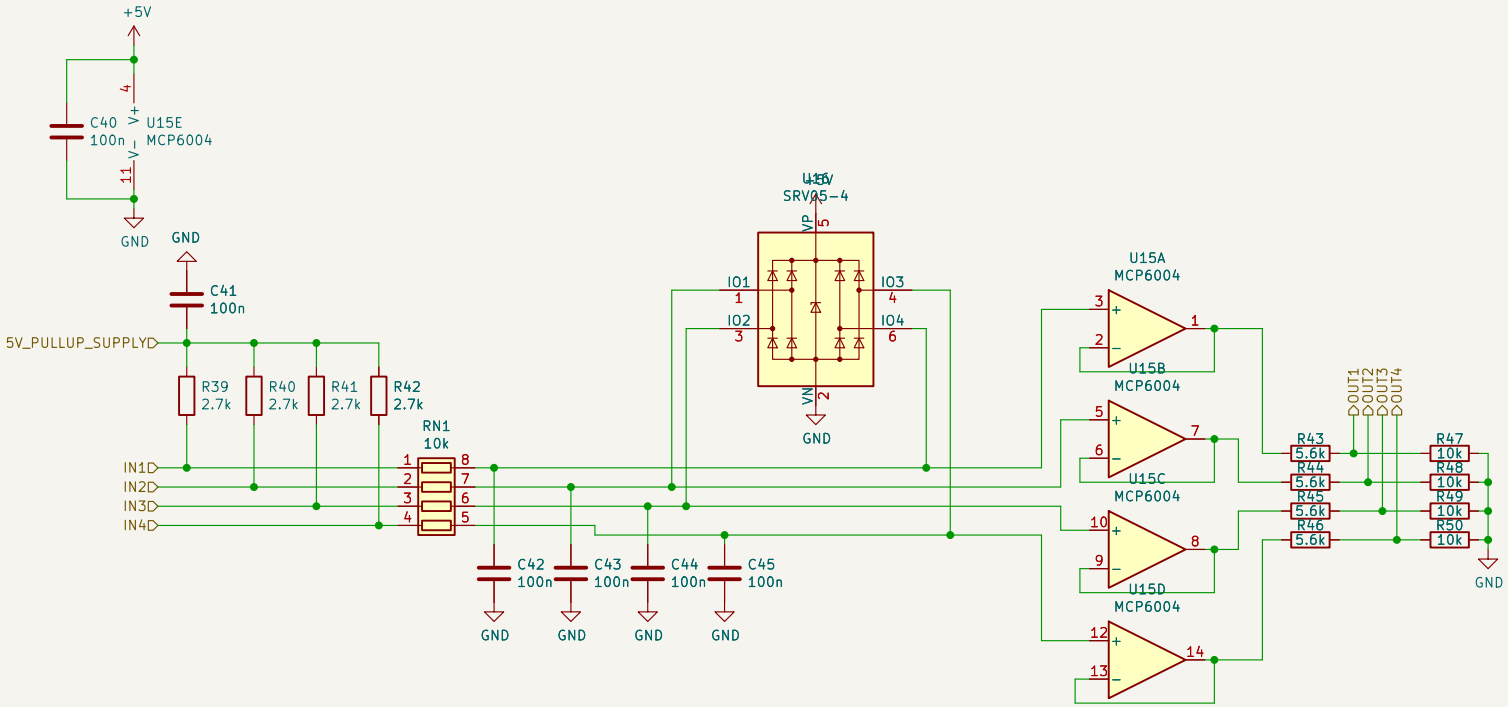
rusefi.com/s/proteus

Sheet: /knock/
File: knock.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 3/21


5V_SENSOR_SUPPLY
R23 2.7k
R29 2.7k
C24 1n
C30 1n
U10A
SN74LVC2G17DB
+5V
HALL_IN_1
HALL_OUT_1
GND
R24 2.7k
R30 2.7k
C25 1n
C31 1n
U8A
HALL_IN_2
HALL_OUT_2
R25 2.7k
R31 2.7k
C26 1n
C32 1n
U8B
HALL_IN_3
HALL_OUT_3
R26 2.7k
R32 2.7k
C27 1n
C33 1n
U9A
HALL_IN_4
HALL_OUT_4
R27 2.7k
R33 2.7k
C28 1n
C34 1n
U10B
HALL_IN_5
HALL_OUT_5
R28 2.7k
R34 2.7k
C29 1n
C35 1n
U9B
HALL_IN_6
HALL_OUT_6
U12
SRV05-4
U11
IO1
IO2
IO3
IO4
VP
VN
74HC parts include internal clamp diodes to the power rails
R36 2.7k
R38 2.7k
C37 1n
C39 1n
U14A
VR1
VR1_OUT
U13
R35 2.7k
R37 2.7k
C36 1n
C38 1n
U14B
VR2
VR2_OUT
rusefi.com/s/proteus
Sheet: /triggers/
File: triggers.kicad_sch
Title: fome-mc
Size: A4
Date: 2024-11-14
Rev: v1.0-atlas-h7
KiCad E.D.A. 10.0.7
Id: 4/21

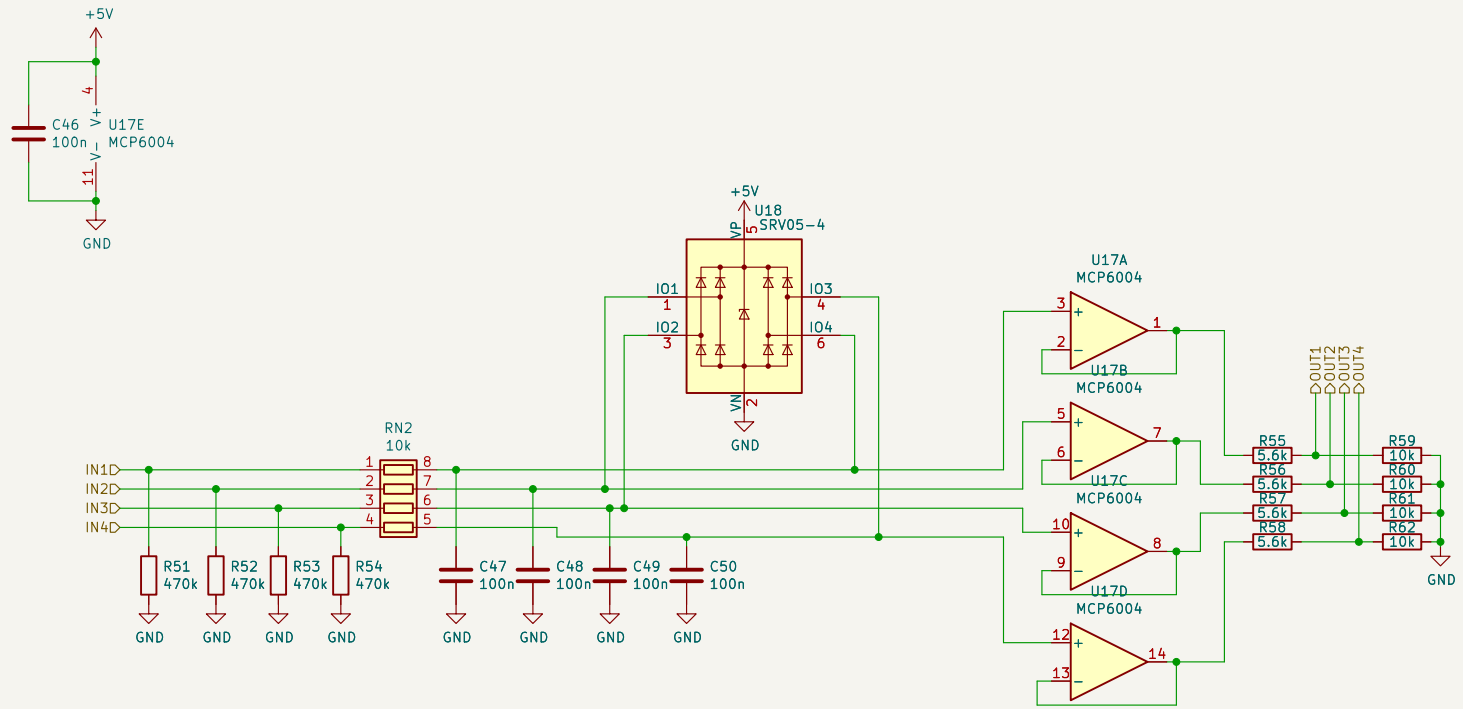


rusefi.com/s/proteus

Sheet: /quad_analog_temp/
File: quad_analog_pullup.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 5/21

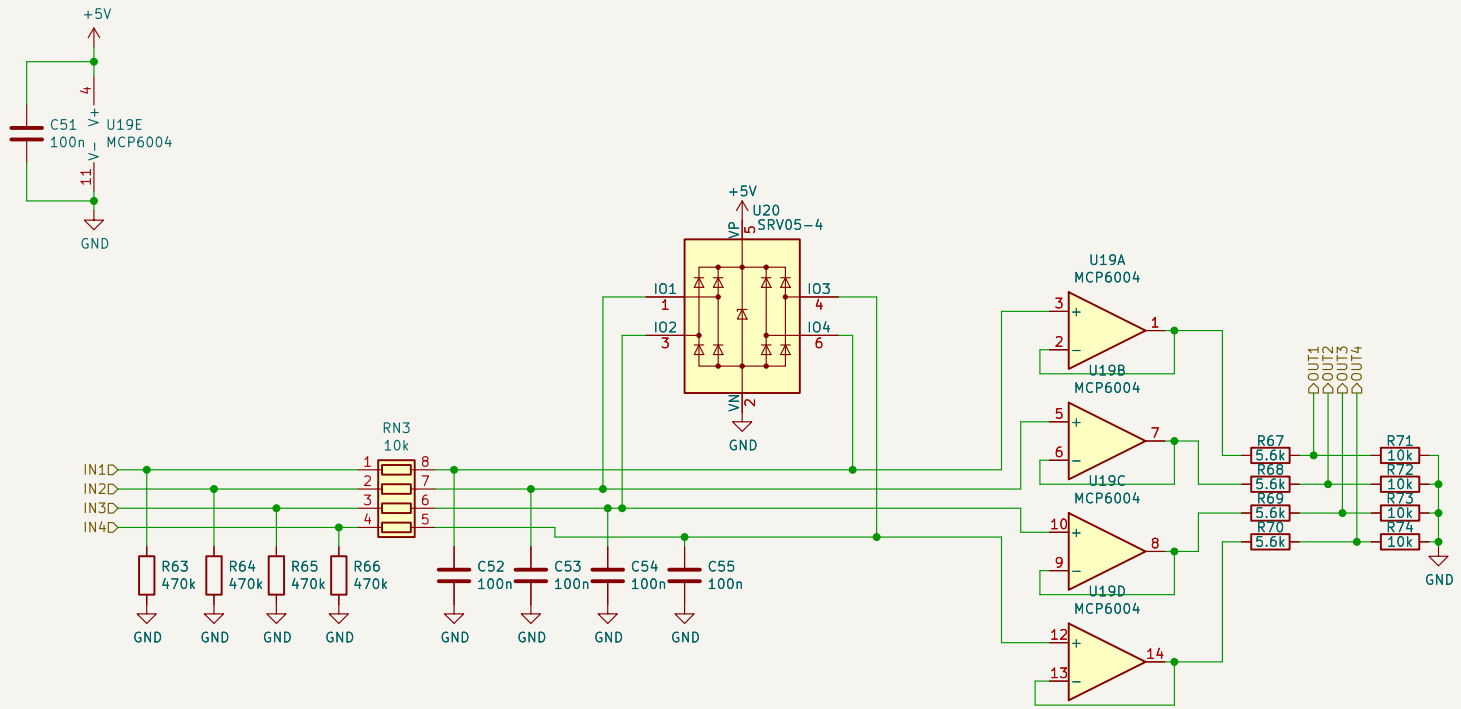


rusefi.com/s/proteus

Sheet: /quad_analog1/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 6/21

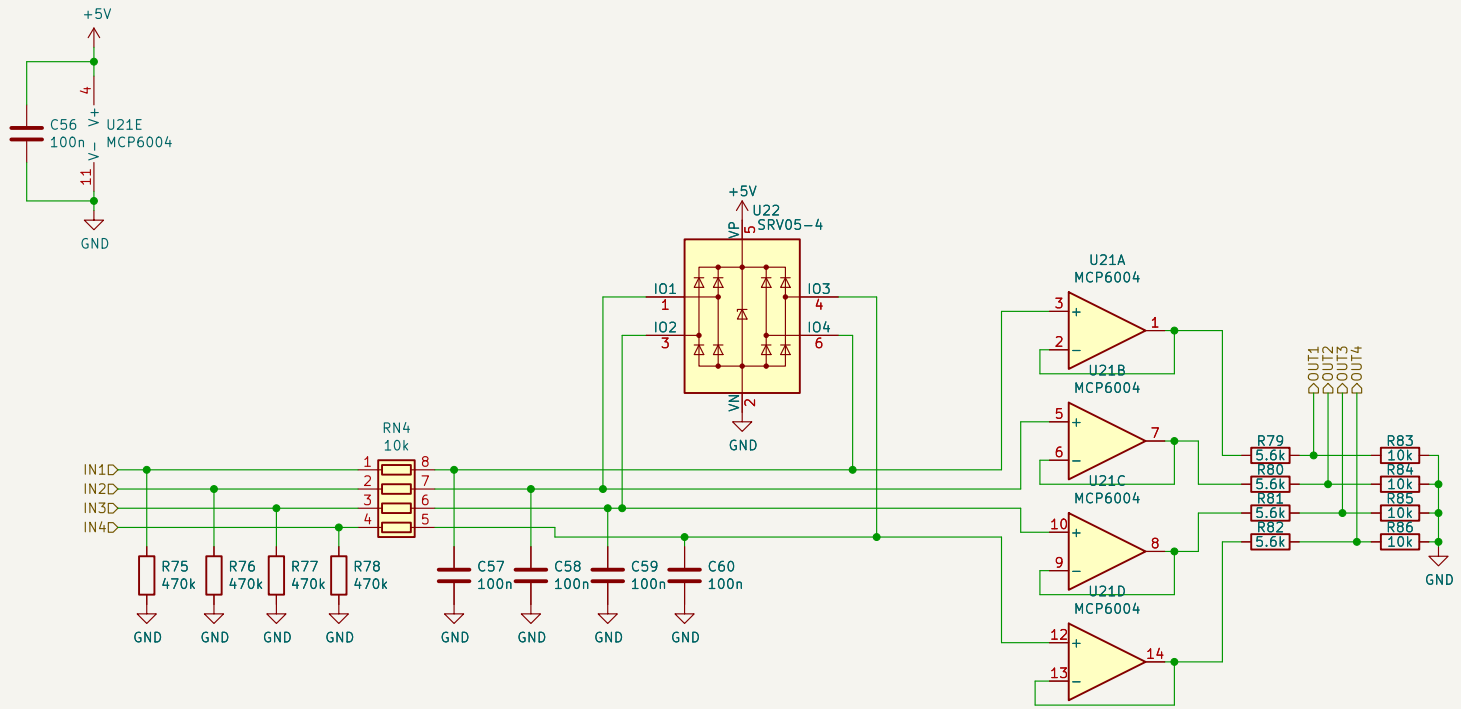


rusefi.com/s/proteus

Sheet: /quad_analog2/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 7/21

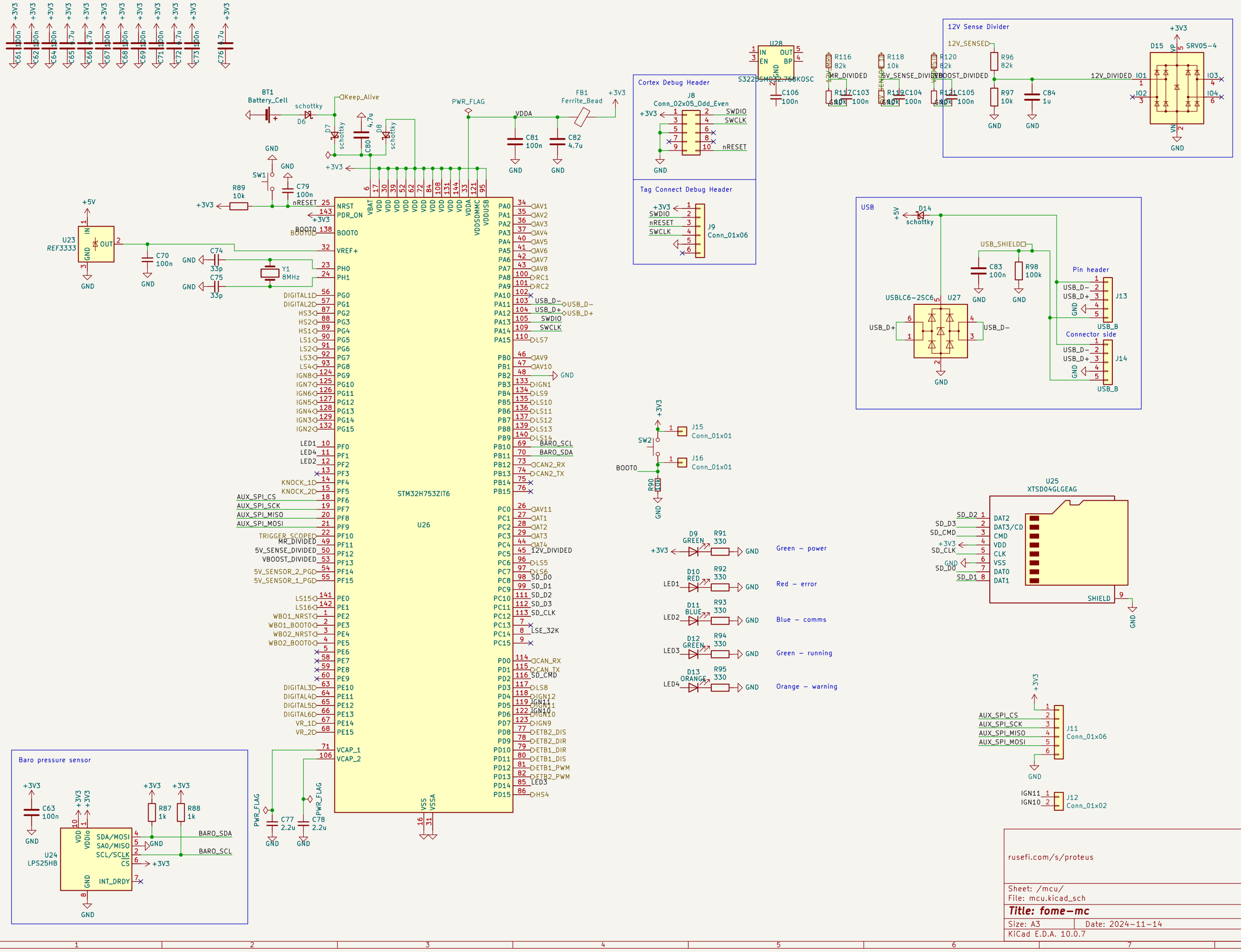


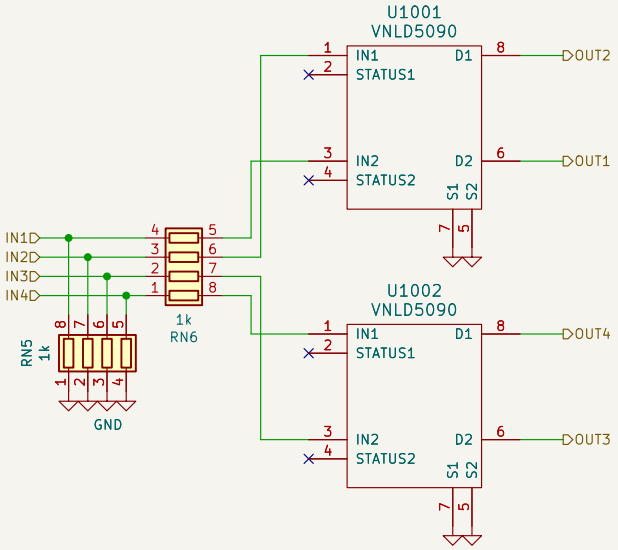
rusefi.com/s/proteus

Sheet: /quad_analog3/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 8/21


STM32H753ZIT6
U26
Cortex Debug Header
J8
Conn_02x05_Odd_Even
Tag Connect Debug Header
J9
Conn_01x06
12V Sense Divider
USB
USBLC6-2SC6
U27
Pin header
Connector side
U25
XTSD04GLGEAG
Baro pressure sensor
U24
LPS25HB
U23
REF3333
U28
S3225SM8032.768KOSC
BT1
Battery_Cell
FB1
Ferrite_Bead
Green – power
Red – error
Blue – comms
Green – running
Orange – warning
rusefi.com/s/proteus
Sheet: /mcu/
File: mcu.kicad_sch
Title: fome–mc
Size: A3
Date: 2024–11–14
Rev: v1.0–atlas–h7
KiCad E.D.A. 10.0.7
Id: 9/21



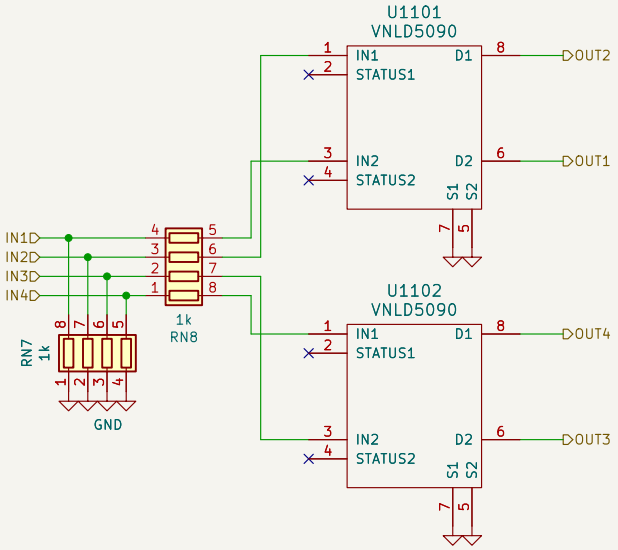
Locate output caps near connector pin

rusefi.com/s/proteus

Sheet: /lowside_quad1/
File: lowside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 10/21



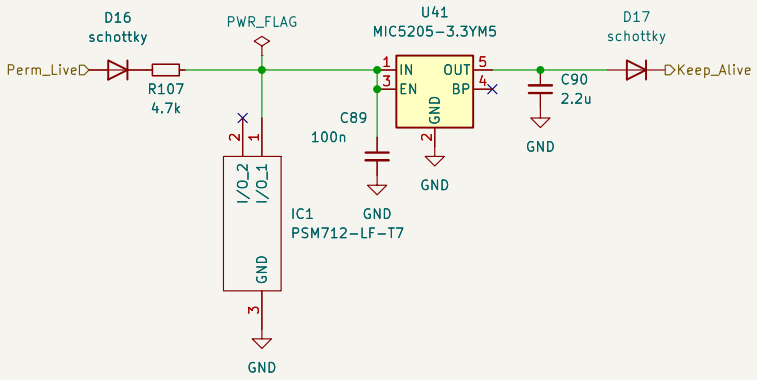
Locate output caps near connector pin

rusefi.com/s/proteus

Sheet: /lowside_quad2/
File: lowside_quad.kicad_sch

Title: fome-mc

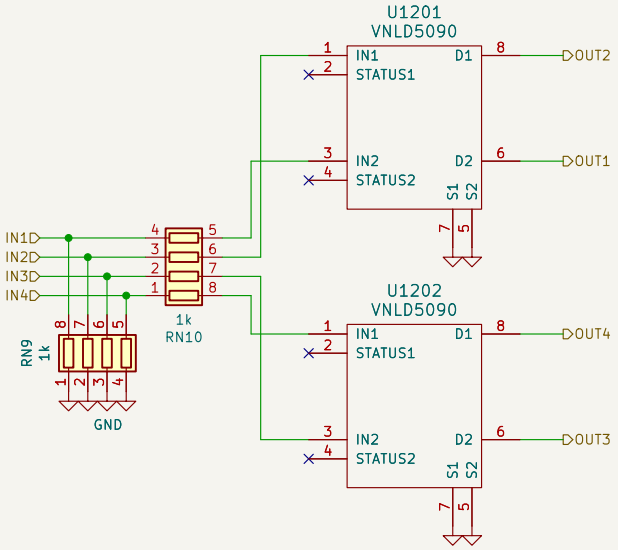
Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 11/21



Sheet: /Keep_Alive/
File: Keep_Alive.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 12/21



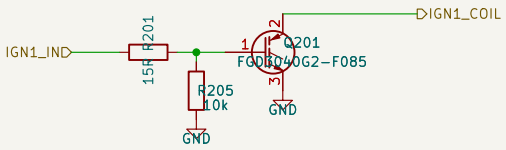
Locate output caps near connector pin

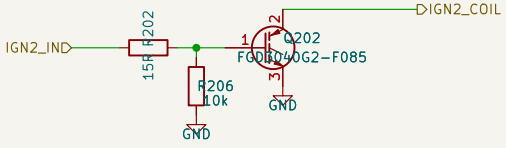
rusefi.com/s/proteus

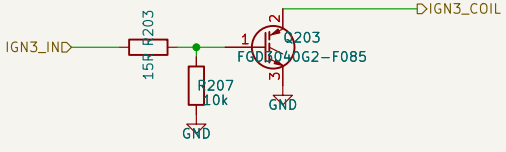
Sheet: /lowside_quad4/
File: lowside_quad.kicad_sch

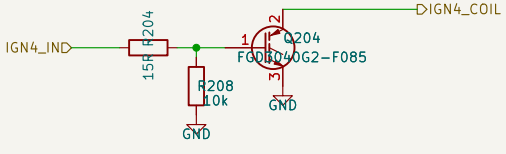
Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 13/21





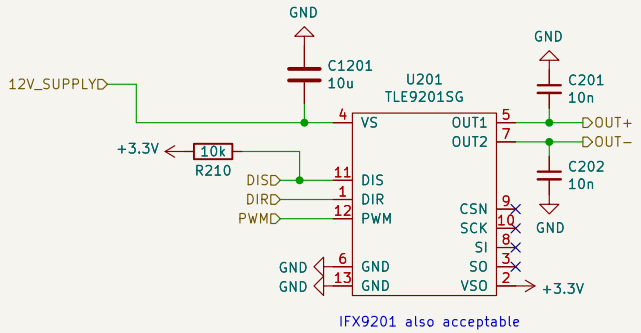




Sheet: /ign_igbt/
File: ign_igbt.kicad_sch

Title: fome-mc 4x High-Current IGBT Ignition Stage

Size: A4	Date: 2026-10-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 14/21



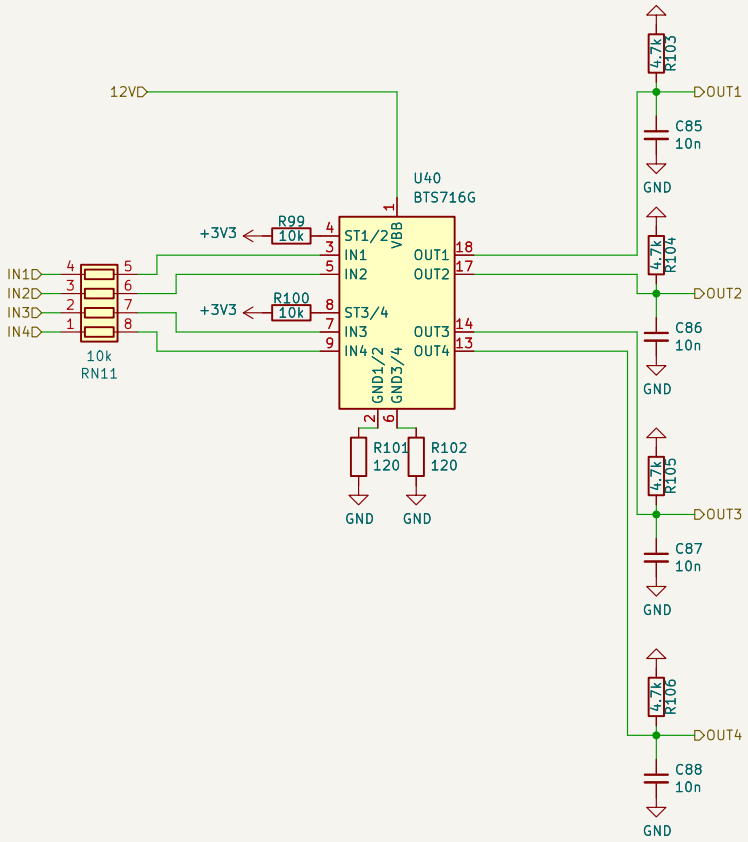
IFX9201 also acceptable

rusefi.com/s/proteus
github.com/mck1117/proteus
rusEFI

Sheet: /etb1/
File: etb-9201.kicad_sch

Title: rusEfi Proteus

Size: A4	Date: 2022-04-09	**Rev: v0.7**
KiCad E.D.A. 10.0.7		Id: 15/21

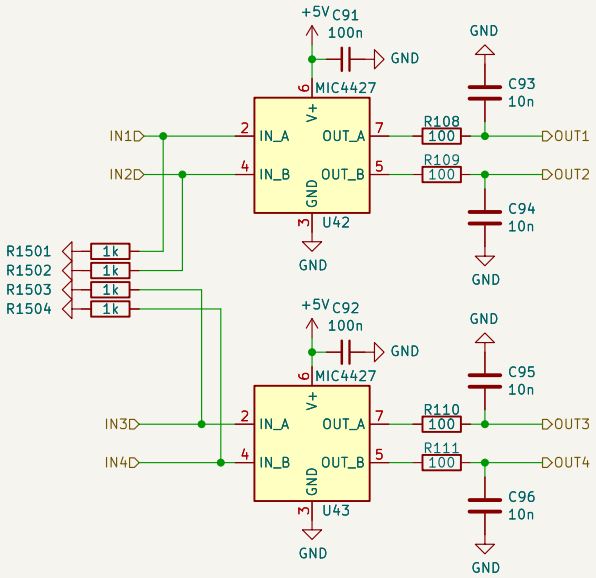


rusefi.com/s/proteus

Sheet: /highside_quad/
File: highside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 16/21

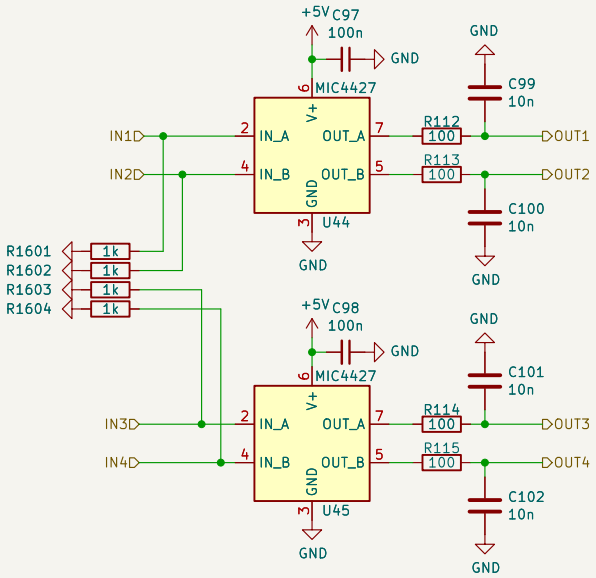


rusefi.com/s/proteus

Sheet: /ign1/
File: ign4.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 17/21

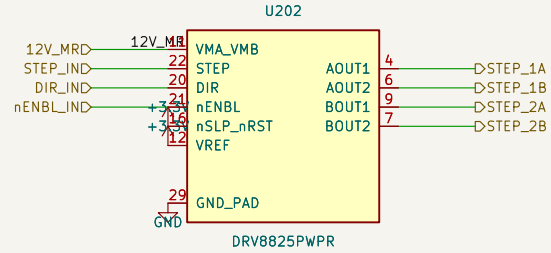


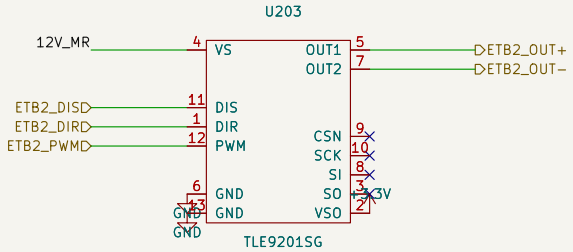
rusefi.com/s/proteus

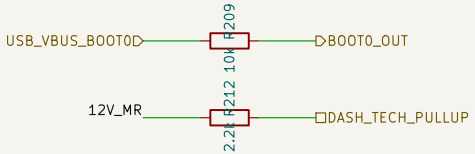
Sheet: /ign2/
File: ign4.kicad_sch

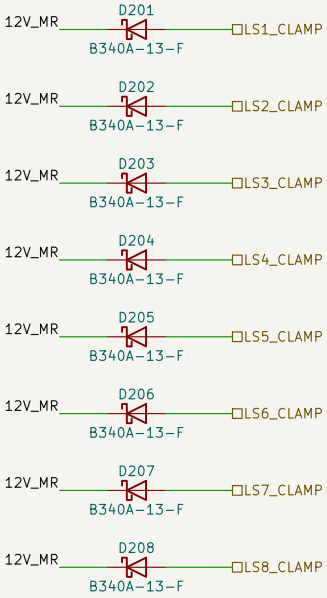
Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 18/21





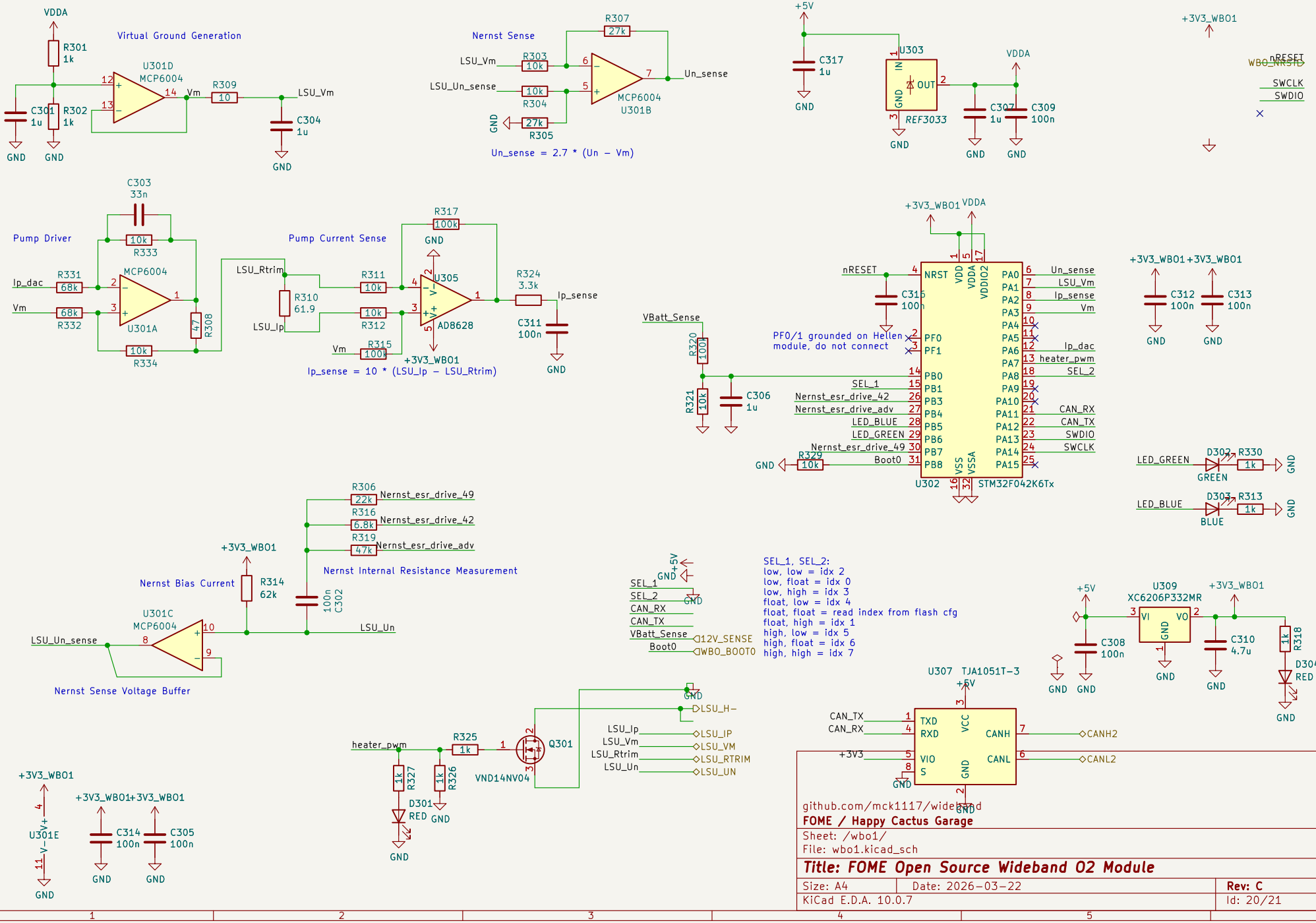


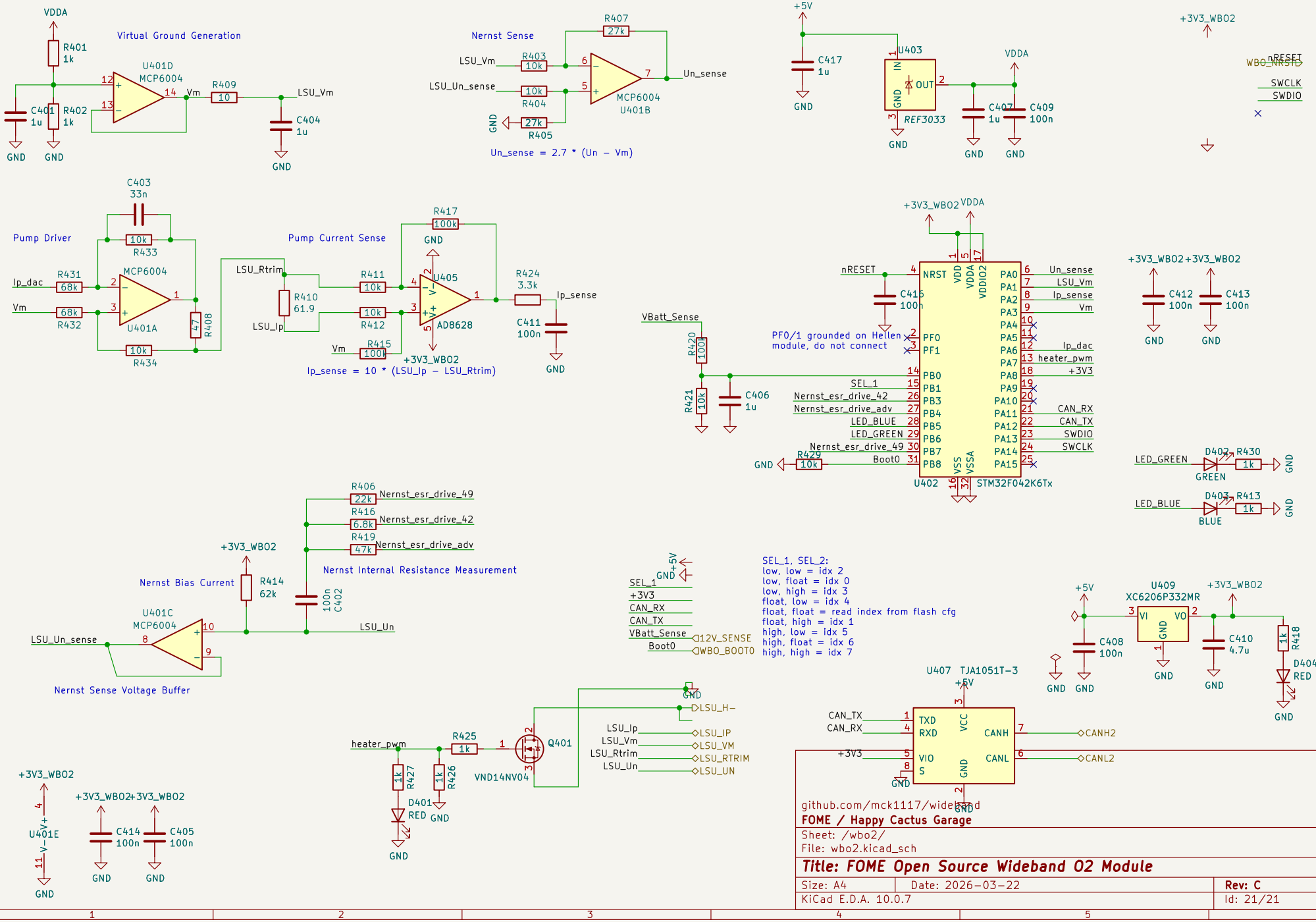


Sheet: /stepper_etb/
File: stepper_etb.kicad_sch

Title: fome-mc Stepper Driver + ETB2 H-Bridge + Low-Side Clamps + BOO

Size: A4	Date: 2026-10-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 19/21


Virtual Ground Generation
VDDA
R301 1k
R302 1k
C301 1u
U301D MCP6004
R309 10
C304 1u
LSU_Vm
Vm
GND
Nernst Sense
R307 27k
R303 10k
R304 10k
R305 27k
LSU_Vm
LSU_Un_sense
Un_sense
MCP6004 U301B
Un_sense = 2.7 * (Un − Vm)
+5V
C317 1u
U303
REF3033
C307 1u
C309 100n
+3V3_WBO1
nRESET
WBO_NRST
SWCLK
SWDIO
C303 33n
Pump Driver
R333 10k
R331 68k
R332 68k
R334 10k
R308 47
U301A MCP6004
Ip_dac
Vm
Pump Current Sense
R317 100k
R310 61.9
R311 10k
R312 10k
R315 100k
U305 AD8628
R324 3.3k
C311 100n
Ip_sense
LSU_Rtrim
LSU_Ip
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
+3V3_WBO1
VDDA
U302 STM32F042K6Tx
nRESET
C316 100n
PF0/1 grounded on Hellen module, do not connect
VBatt_Sense
R320 100k
R321 10k
C306 1u
R329 10k
Boot0
SEL_1
Nernst_esr_drive_42
Nernst_esr_drive_adv
LED_BLUE
LED_GREEN
Nernst_esr_drive_49
Un_sense
LSU_Vm
Ip_sense
Vm
Ip_dac
heater_pwm
SEL_2
CAN_RX
CAN_TX
SWDIO
SWCLK
C312 100n
C313 100n
LED_GREEN
D302 GREEN
R330 1k
LED_BLUE
D303 BLUE
R313 1k
R306 22k
R316 6.8k
R319 47k
Nernst_esr_drive_49
Nernst_esr_drive_42
Nernst_esr_drive_adv
Nernst Internal Resistance Measurement
Nernst Bias Current
R314 62k
C302 100n
U301C MCP6004
LSU_Un_sense
LSU_Un
Nernst Sense Voltage Buffer
SEL_1
SEL_2
CAN_RX
CAN_TX
VBatt_Sense
Boot0
12V_SENSE
WBO_BOOT0
SEL_1, SEL_2:
low, low = idx 2
low, float = idx 0
low, high = idx 3
float, low = idx 4
float, float = read index from flash cfg
float, high = idx 1
high, low = idx 5
high, float = idx 6
high, high = idx 7
U309 XC6206P332MR
C308 100n
C310 4.7u
R318 1k
D304 RED
U307 TJA1051T-3
CAN_TX
CAN_RX
+3V3
CANH
CANL
CANH2
CANL2
heater_pwm
R325 1k
Q301 VND14NV04
R327 1k
R326 1k
D301 RED
LSU_Ip
LSU_Vm
LSU_Rtrim
LSU_Un
LSU_IP
LSU_VM
LSU_RTRIM
LSU_UN
LSU_H−
U301E
C314 100n
C305 100n
github.com/mck1117/widebond
FOME / Happy Cactus Garage
Sheet: /wbo1/
File: wbo1.kicad_sch
Title: FOME Open Source Wideband O2 Module
Size: A4
Date: 2026-03-22
Rev: C
KiCad E.D.A. 10.0.7
Id: 20/21


Virtual Ground Generation
VDDA
R401 1k
C401 1u
R402 1k
U401D MCP6004
R409 10
LSU_Vm
C404 1u
GND
Nernst Sense
R407 27k
R403 10k
LSU_Vm
LSU_Un_sense
R404 10k
R405 27k
MCP6004 U401B
Un_sense
Un_sense = 2.7 * (Un − Vm)
+5V
C417 1u
U403 REF3033
VDDA
C407 1u
C409 100n
+3V3_WBO2
nRESET
WBO_NRST
SWCLK
SWDIO
Pump Driver
C403 33n
R433 10k
Ip_dac
R431 68k
Vm
R432 68k
U401A MCP6004
R434 10k
R408 47
Pump Current Sense
R417 100k
LSU_Rtrim
R410 61.9
LSU_Ip
R411 10k
R412 10k
R415 100k
U405 AD8628
+3V3_WBO2
R424 3.3k
Ip_sense
C411 100n
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
+3V3_WBO2 VDDA
nRESET
C416 100n
PF0/1 grounded on Hellen module, do not connect
VBatt_Sense
R420 100k
R421 10k
C406 1u
SEL_1
Nernst_esr_drive_42
Nernst_esr_drive_adv
LED_BLUE
LED_GREEN
Nernst_esr_drive_49
R429 10k
Boot0
U402 STM32F042K6Tx
Un_sense
LSU_Vm
Ip_sense
Vm
Ip_dac
heater_pwm
+3V3
CAN_RX
CAN_TX
SWDIO
SWCLK
+3V3_WBO2 +3V3_WBO2
C412 100n
C413 100n
LED_GREEN
D402 GREEN
R430 1k
LED_BLUE
D403 BLUE
R413 1k
R406 22k
Nernst_esr_drive_49
R416 6.8k
Nernst_esr_drive_42
R419 47k
Nernst_esr_drive_adv
Nernst Internal Resistance Measurement
+3V3_WBO2
Nernst Bias Current
R414 62k
C402 100n
U401C MCP6004
LSU_Un_sense
LSU_Un
Nernst Sense Voltage Buffer
SEL_1
+3V3
CAN_RX
CAN_TX
VBatt_Sense
Boot0
12V_SENSE
WBO_BOOT0
SEL_1, SEL_2:
low, low = idx 2
low, float = idx 0
low, high = idx 3
float, low = idx 4
float, float = read index from flash cfg
float, high = idx 1
high, low = idx 5
high, float = idx 6
high, high = idx 7
+5V
U409 XC6206P332MR
C408 100n
C410 4.7u
R418 1k
D404 RED
U407 TJA1051T-3
CAN_TX
CAN_RX
+3V3
CANH
CANL
CANH2
CANL2
LSU_H−
LSU_Ip
LSU_Vm
LSU_Rtrim
LSU_Un
LSU_IP
LSU_VM
LSU_RTRIM
LSU_UN
heater_pwm
R425 1k
Q401 VND14NV04
R427 1k
R426 1k
D401 RED
+3V3_WBO2
U401E
+3V3_WBO2 +3V3_WBO2
C414 100n
C405 100n
github.com/mck1117/widebond
FOME / Happy Cactus Garage
Sheet: /wbo2/
File: wbo2.kicad_sch
Title: FOME Open Source Wideband O2 Module
Size: A4
Date: 2026-03-22
Rev: C
KiCad E.D.A. 10.0.7
Id: 21/21